B.M.S COLLEGE OF ENGINEERING BENGALURU
Autonomous Institute, Affiliated to VTU

AAT Report on

Design and verification of Digital circuits using Cadence Tool

Submitted in partial fulfilment of AAT
for the course on

Advanced Digital Logic Design

By,
VIGNESH
1BM24EC249

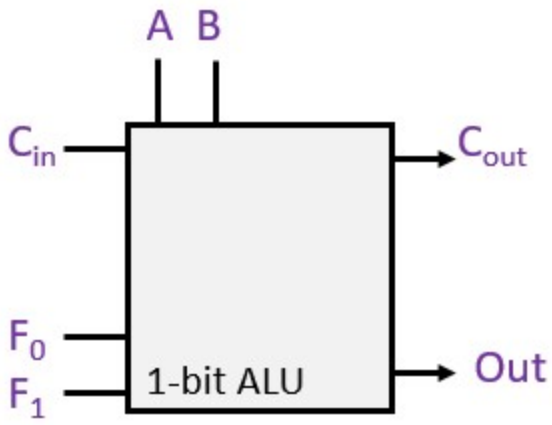
To **Prof. Ajaykumar Devarapalli**
Assistant Professor

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2025-2026

Arithmetic Logic Unit

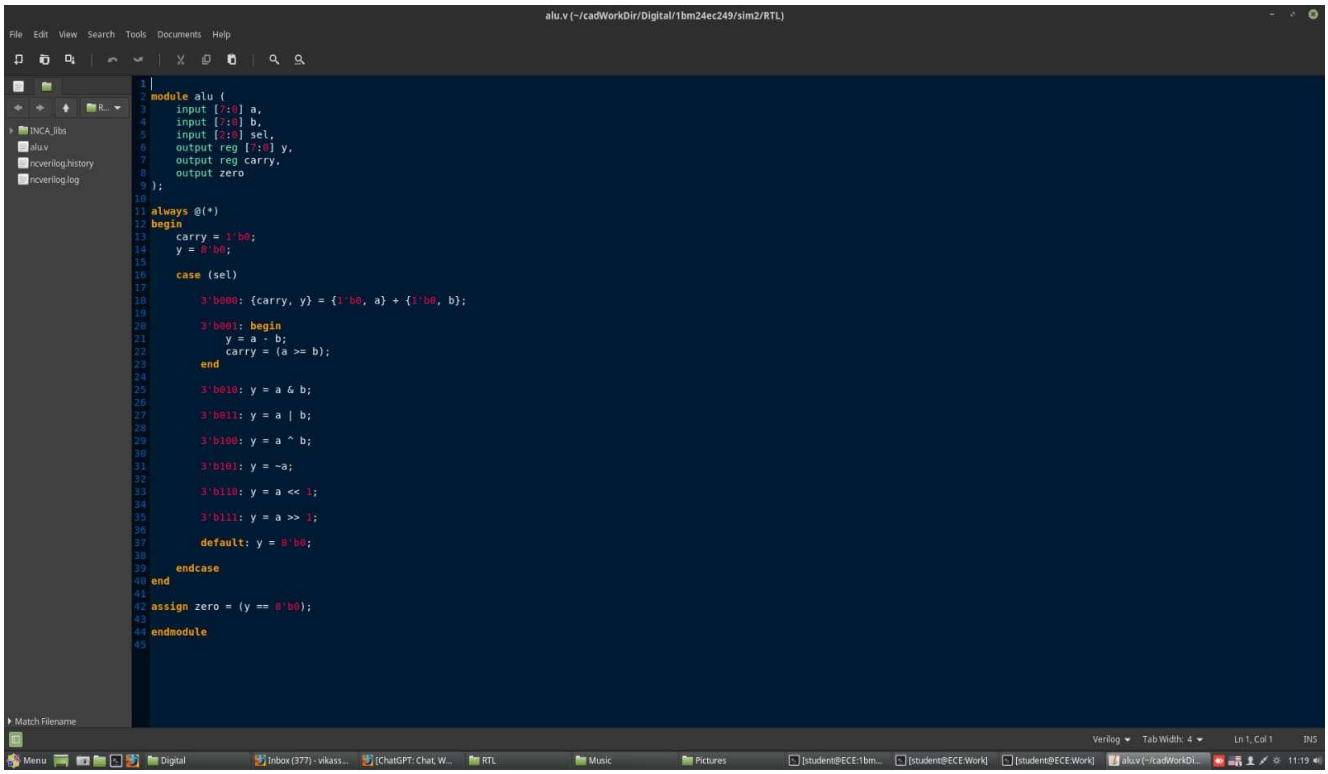
Introduction To ALU:



The Arithmetic Logic Unit (ALU) is one of the most important components of a digital computer. It is a part of the Central Processing Unit (CPU) responsible for performing arithmetic and logical operations on binary data. The ALU plays a crucial role in processing instructions and executing calculations required by various computer applications.

An ALU performs arithmetic operations such as addition, subtraction, multiplication, and division, along with logical operations such as AND, OR, NOT, and XOR. It can also perform comparison and bitwise operations depending on its design.

CODE:




```
module alu (
    input [7:0] a,
    input [7:0] b,
    input [2:0] sel,
    output reg [7:0] y,
    output reg carry,
    output zero
);

always @(*)
begin
    carry = 1'b0;
    y = 8'b0;

    case (sel)

        3'b000: {carry, y} = {1'b0, a} + {1'b0, b};

        3'b001: begin
            y = a - b;
            carry = (a >= b);
        end

        3'b010: y = a & b;

        3'b011: y = a | b;

        3'b100: y = a ^ b;

        3'b101: y = ~a;

        3'b110: y = a << 1;

        3'b111: y = a >> 1;

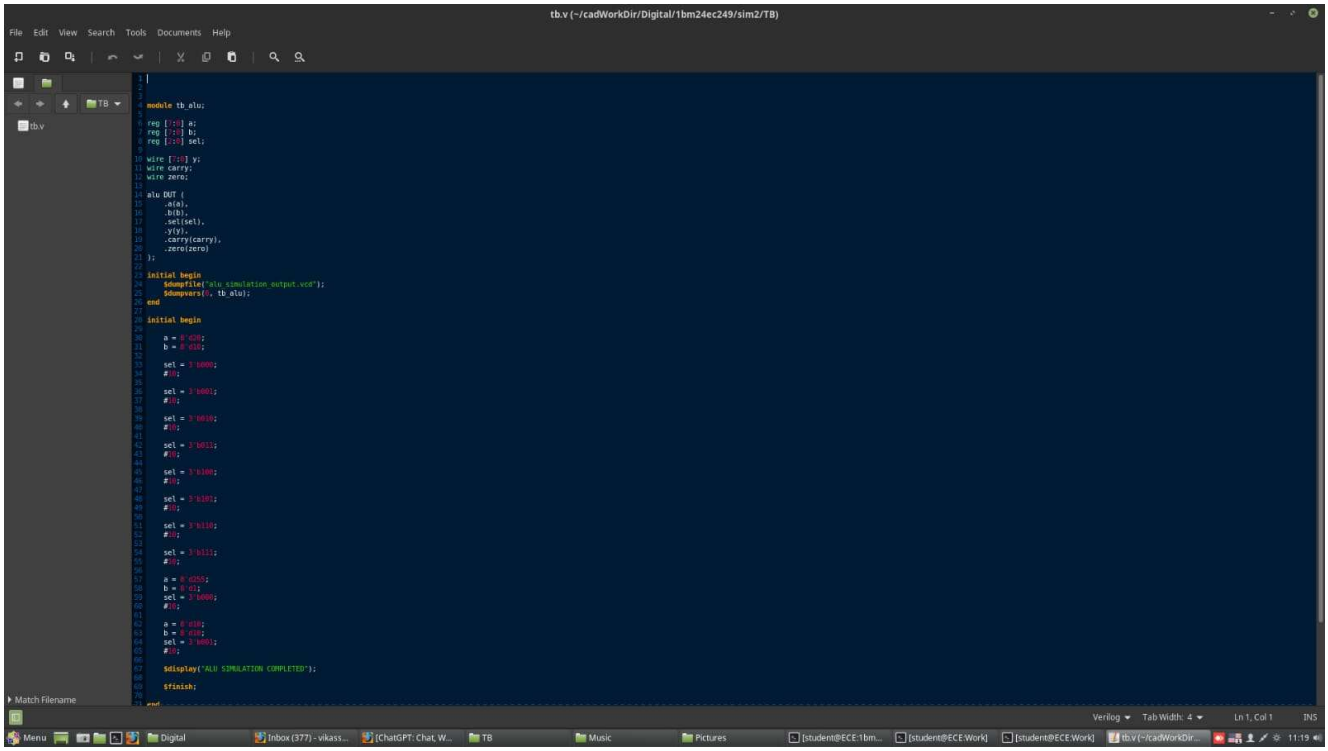
        default: y = 8'b0;

    endcase
end

assign zero = (y == 8'b0);

endmodule
```

TESTBENCH:




```
module tb_alu;

reg [7:0] a;
reg [7:0] b;
reg [2:0] sel;

wire [7:0] y;
wire carry;
wire zero;

alu DUT (
    .a(a),
    .b(b),
    .sel(sel),
    .y(y),
    .carry(carry),
    .zero(zero)
);

initial begin
    $dumpfile("alu_simulation_output.vcd");
    $dumpvars(0, tb_alu);
end

initial begin

    a = 8'd20;
    b = 8'd10;

    sel = 3'b000;
    #10;

    sel = 3'b001;
    #10;

    sel = 3'b010;
    #10;

    sel = 3'b011;
    #10;

    sel = 3'b100;
    #10;

    sel = 3'b101;
    #10;

    sel = 3'b110;
    #10;

    sel = 3'b111;
    #10;

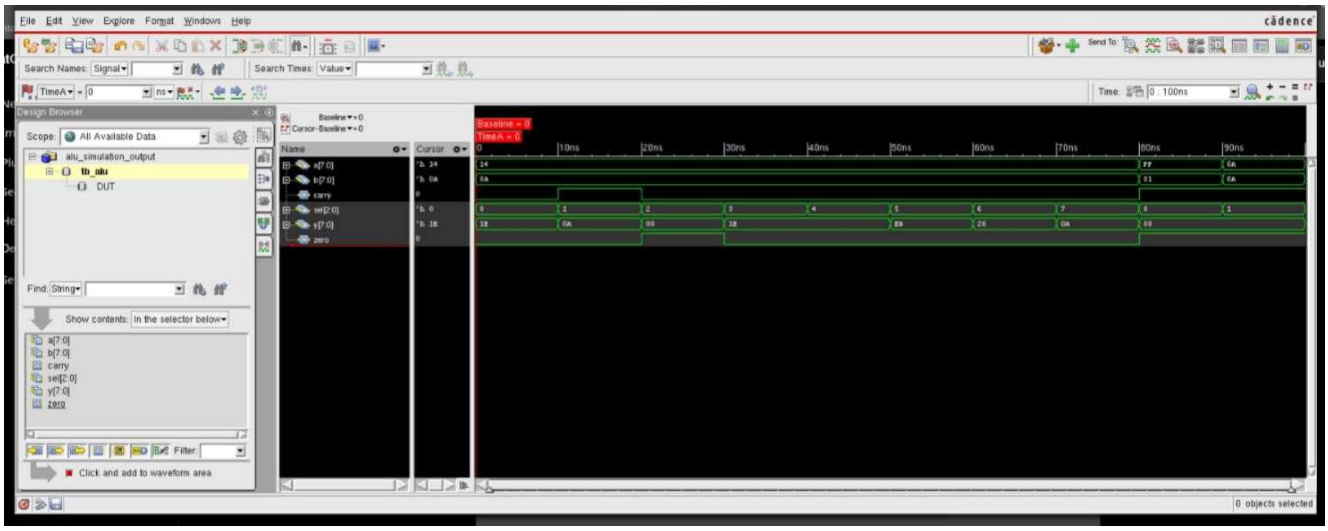
    a = 8'd255;
    b = 8'd1;
    sel = 3'b000;
    #10;

    a = 8'd10;
    b = 8'd20;
    sel = 3'b001;
    #10;

    $display("ALU SIMULATION COMPLETED");

    $finish;
end
```

WAVEFORM:



DIVIDE BY N COUNTER

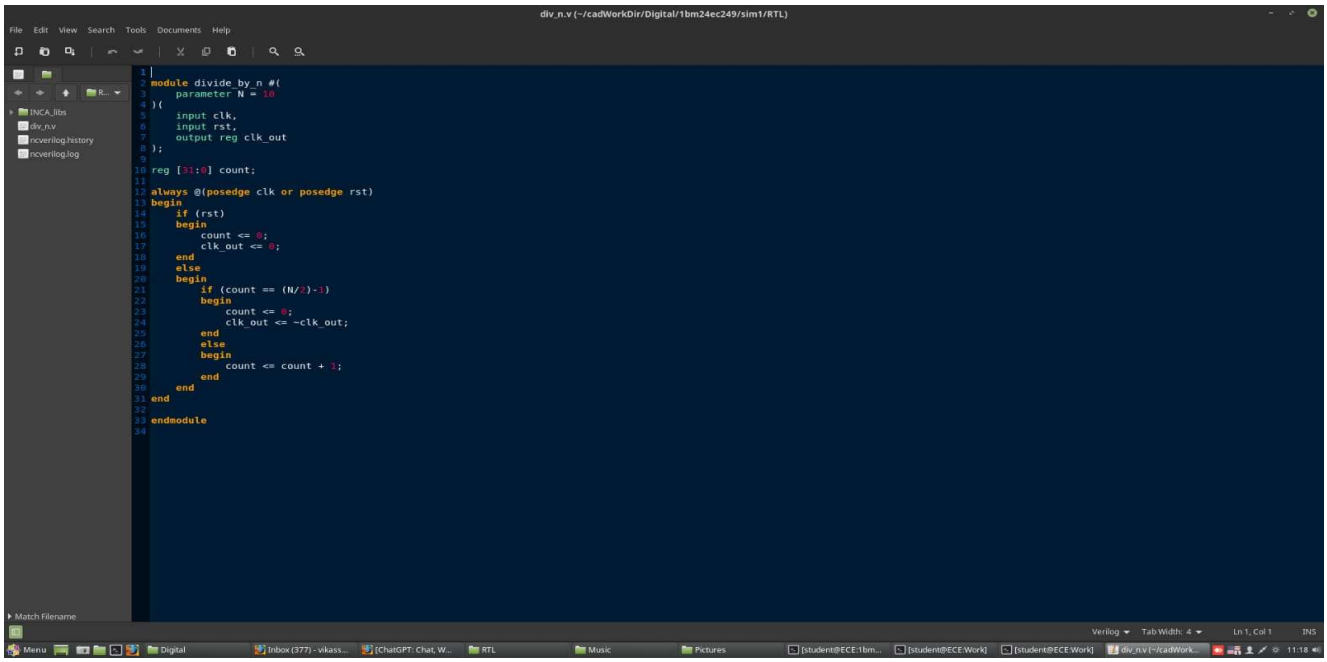
INTRODUCTION TO DIVIDE BY N COUNTER:

A Divide-by-N Counter is an important digital circuit used in digital electronics and communication systems. It is a type of counter that divides the input clock frequency by a specified value, N. The counter counts a fixed number of clock pulses and then resets or repeats its counting sequence, generating an output signal with a lower frequency than the input clock.

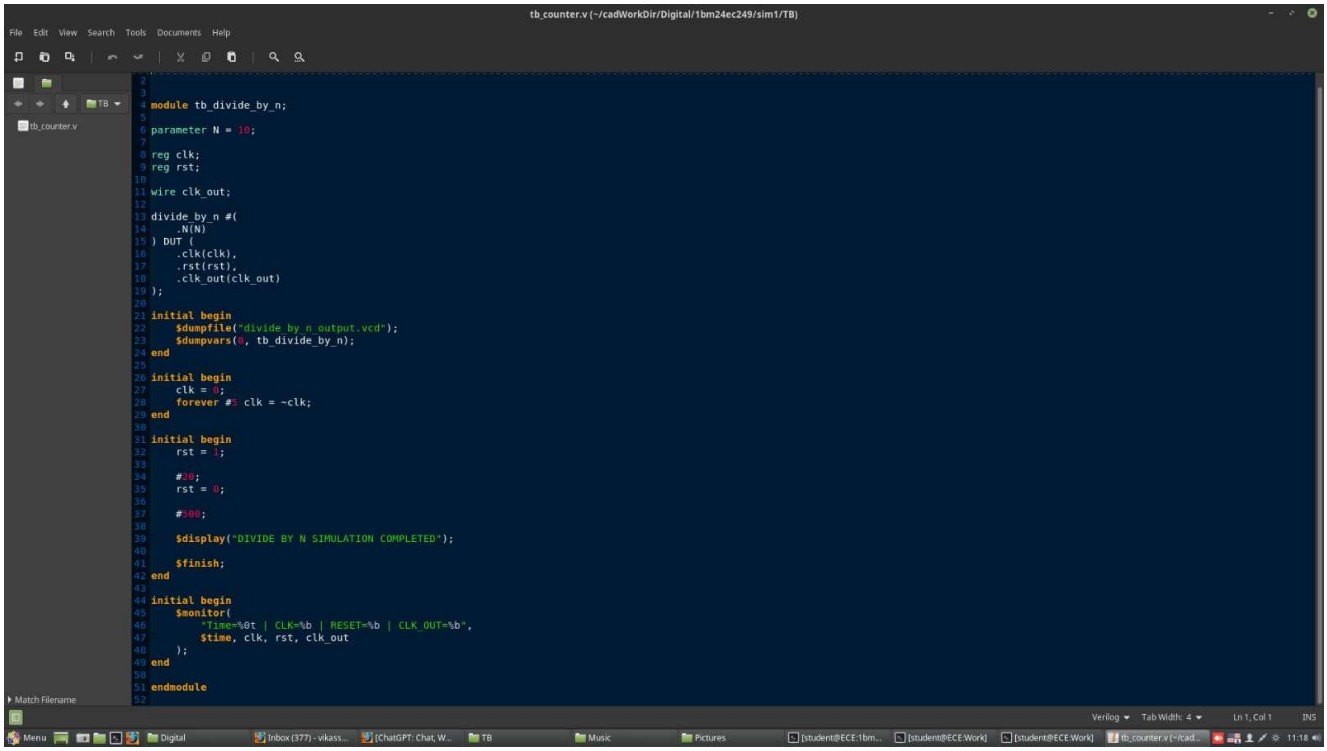
A Divide-by-N Counter is designed using flip-flops and combinational logic circuits. The value of N determines the number of clock pulses required to complete one counting cycle. For example, a Divide-by-10 Counter counts from 0 to 9 and then returns to 0, producing an output frequency that is one-tenth of the input clock frequency.

The main objective of this project is to understand the working principle of counters, frequency division, and sequential logic circuits. It also provides practical knowledge of designing and simulating digital circuits using hardware description languages.

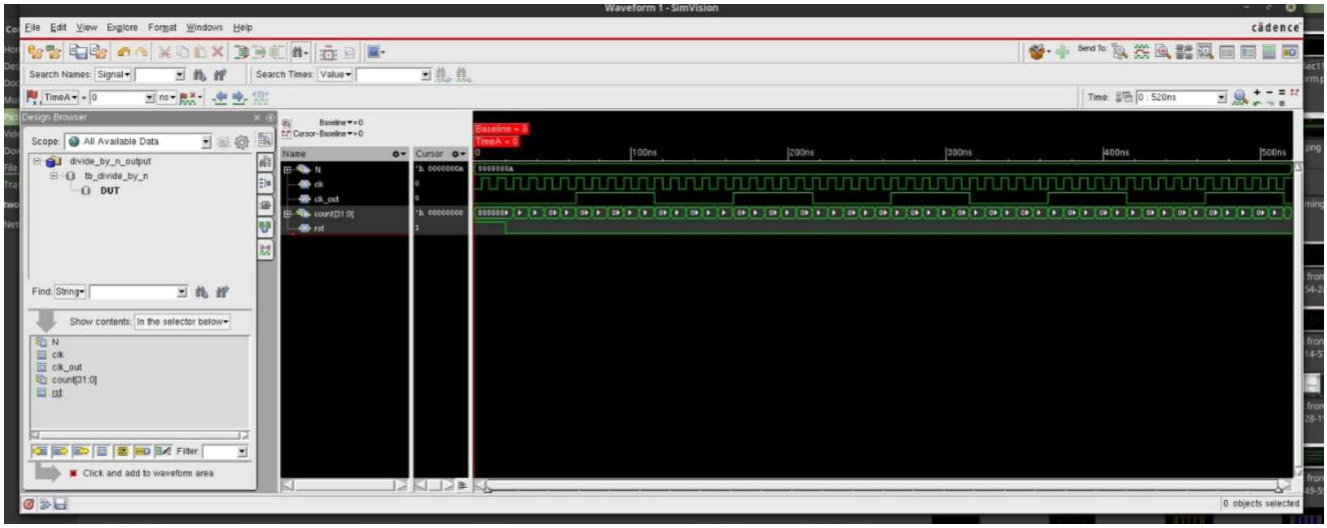
CODE:



TESTBENCH:



WAVEFORM:



GITHUB : https://github.com/naikvignesh45-ai/ADLD-AAT2/upload/main